SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA0716**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 **Total Marks:40**

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS)**.** Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

SOL: **LAN Using Switch – Ethernet Frame Analysis**

Aim

To design a simple LAN with two PCs connected through a switch, configure IP addresses, verify connectivity using ping, and analyze Ethernet frames using Wireshark.

Topology

PC0 -------- Switch -------- PC1

IP Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Procedure

1. Open Cisco Packet Tracer.
2. Place two PCs and one Switch.
3. Connect using Copper Straight-through cable.
4. Configure the IP addresses.
5. Open Command Prompt on PC0.
6. Execute:

    ping 192.168.1.2

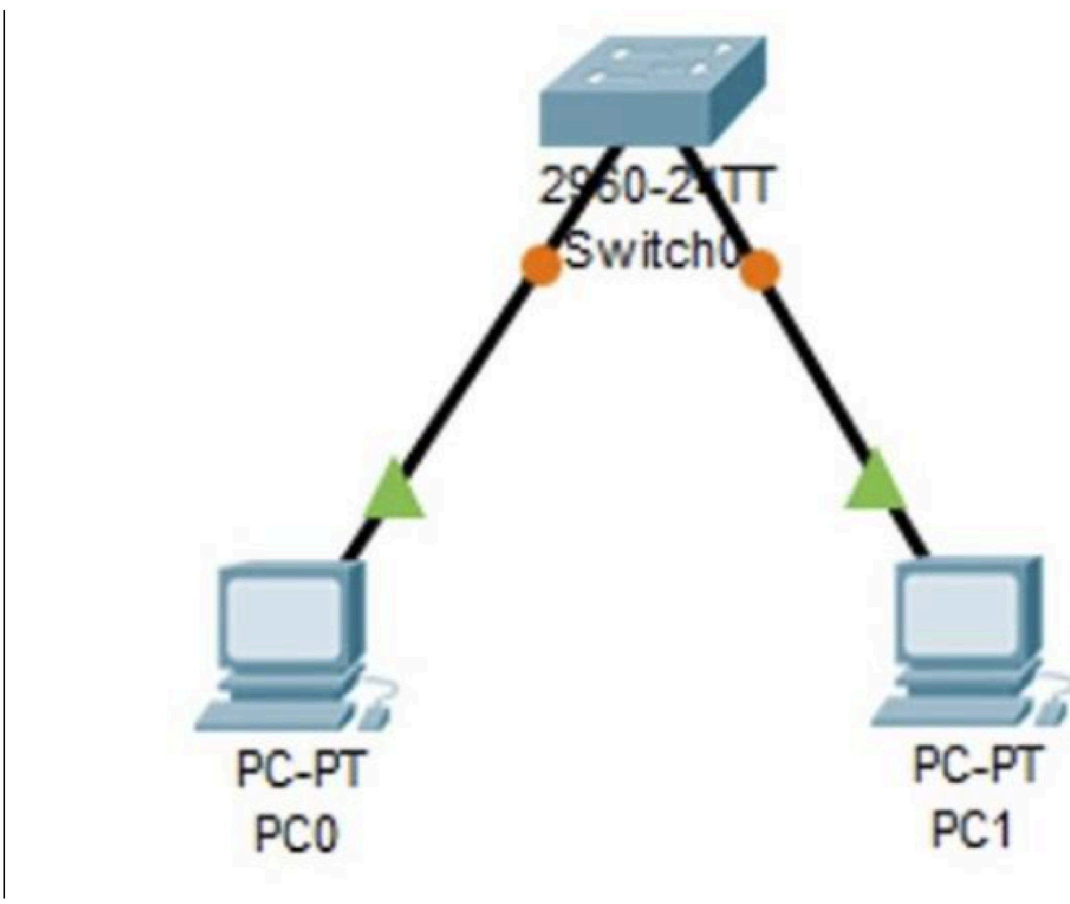
7. Start Wireshark capture.
8. Observe Ethernet frames.

Ethernet Frame Fields

- **Source MAC Address:** Sender's MAC address
- **Destination MAC Address:** Receiver's MAC address
- **EtherType:** Indicates protocol (IPv4 = 0x0800)
- **Frame Check Sequence (FCS):** Detects transmission errors

Result

Ping was successful and Ethernet frame fields were captured successfully.



Explanation

The **Data Link Layer** uses MAC addresses to uniquely identify devices within the LAN. The switch forwards Ethernet frames based on destination MAC addresses, ensuring correct local delivery.

2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer**.** Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

Two LANs Connected Through Router – IPv4 Header Analysis

Aim

To connect two LANs using a router, configure IPv4 addressing, verify communication, and analyze IPv4 packets.

Topology

PC0 --- Switch --- Router --- Switch --- PC1

IP Configuration

Device	IP Address	Mask	Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
Router G0/0	192.168.1.1	255.255.255.0	-
Router G0/1	192.168.2.1	255.255.255.0	-
PC1	192.168.2.2	255.255.255.0	192.168.2.1

Procedure

1. Create the topology.
2. Configure router interfaces.
3. Configure PCs.
4. Ping from PC0:

   ping 192.168.2.2

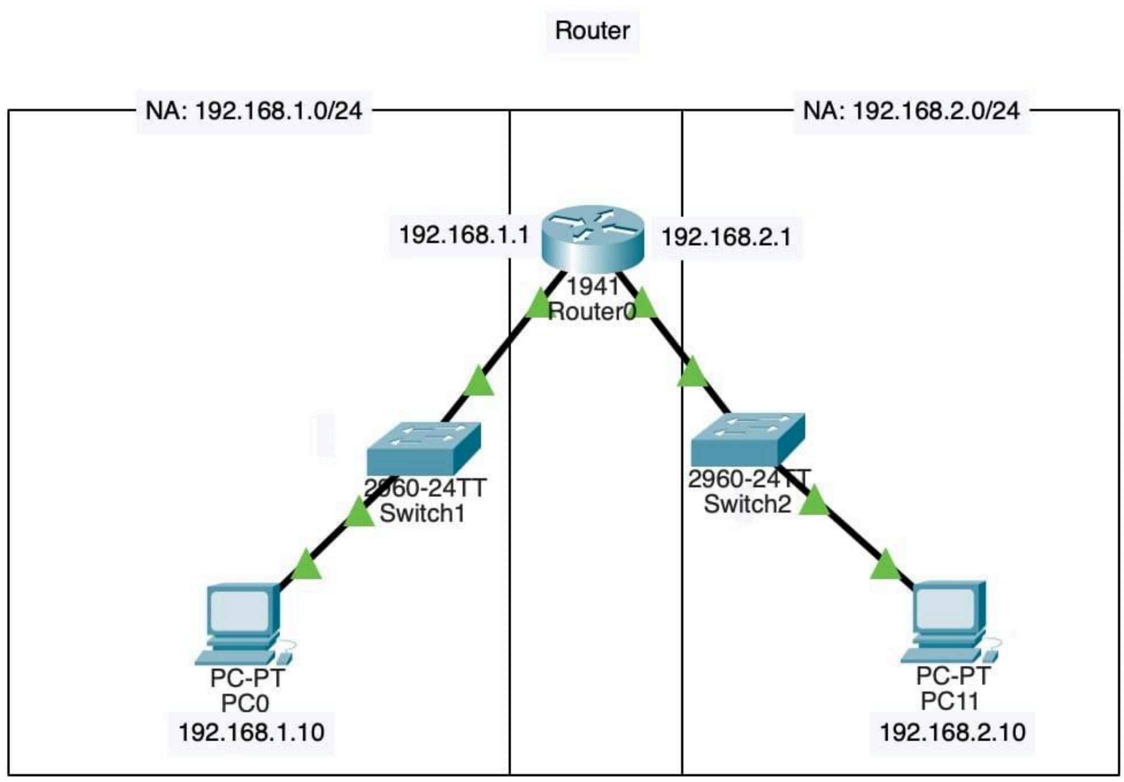
5. Capture packets using Wireshark.

IPv4 Header Fields

- **Source IP Address:** 192.168.1.2
- **Destination IP Address:** 192.168.2.2
- **TTL:** 128 (decreases by 1 at router)
- **Protocol Number:** 1 (ICMP)
- **Header Length:** 20 Bytes

Result

Successful communication between different networks.



Explanation

The **Network Layer** routes packets between different networks using IP addresses. Routers examine destination IP addresses and forward packets to the correct network.

3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

ARP Protocol Analysis

Aim

To study ARP Request and Reply packets and understand IP-to-MAC address resolution.

Topology

PC0 -------- Switch -------- PC1

IP Configuration

Device	IP Address
PC0	192.168.1.10
PC1	192.168.1.20

Procedure

1. Build the LAN.
2. Clear ARP cache.
3. Start Wireshark.
4. Ping PC1 from PC0.
5. Observe ARP packets.

ARP Packets

ARP Request

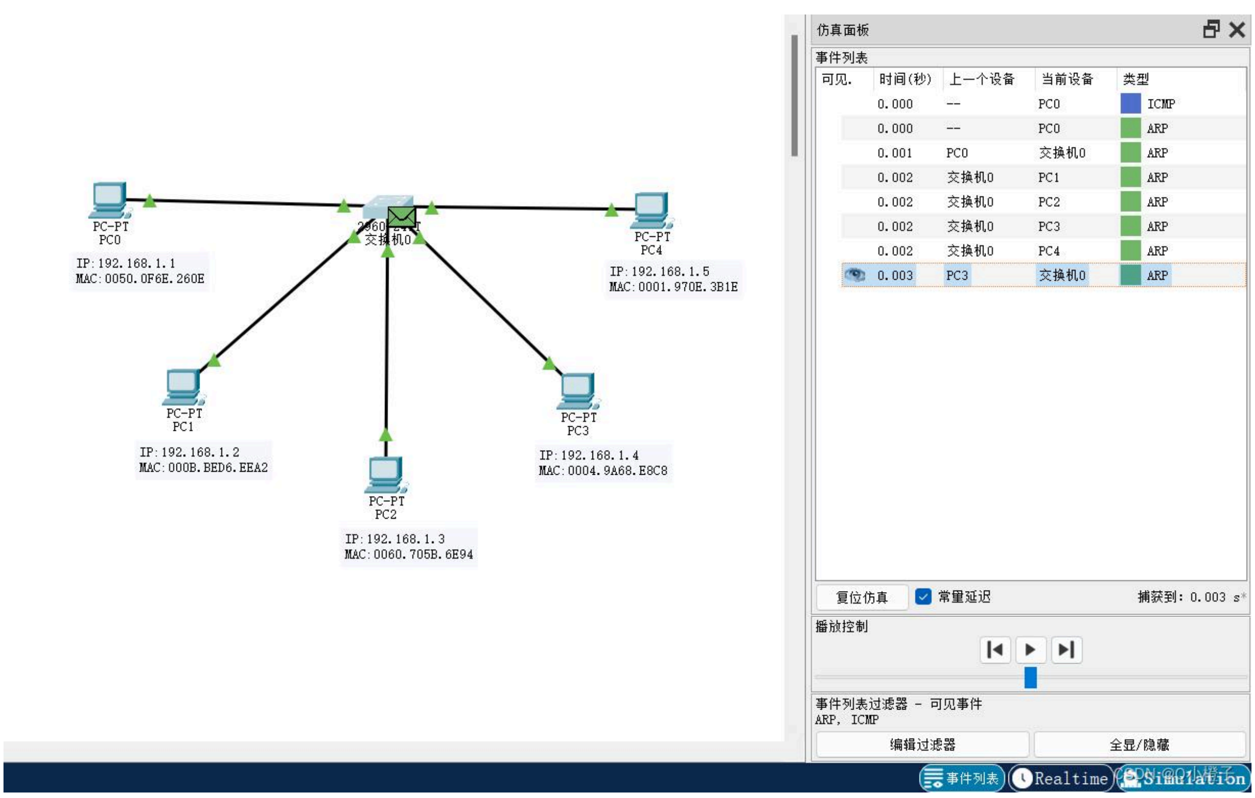
- Sender MAC
- Sender IP
- Target MAC = 00:00:00:00:00:00
- Target IP = 192.168.1.20

ARP Reply

- Sender MAC = PC1 MAC
- Sender IP = 192.168.1.20
- Target MAC = PC0 MAC
- Target IP = 192.168.1.10

Result

ARP Request and Reply packets were successfully captured.



Explanation

ARP converts an IP address into a MAC address before communication. The sender broadcasts an ARP Request, and the destination replies with its MAC address.

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity

using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

SOL:ICMP Packet Analysis

Aim

To capture ICMP Echo Request and Echo Reply messages and analyze their fields.

Topology

PC0 -------- Router -------- PC1

IP Configuration

Device	IP Address
PC0	192.168.1.2
Router G0/0	192.168.1.1
Router G0/1	192.168.2.1
PC1	192.168.2.2

Procedure

1. Configure the network.
2. Ping:

   ping 192.168.2.2

3. Capture packets in Wireshark.

ICMP Fields

- **Type 8** – Echo Request
- **Type 0** – Echo Reply
- **Code** – 0
- **Identifier** – Used to match request and reply
- **Sequence Number** – Tracks packets

Round Trip Time (RTT)

Formula:

RTT = Reply Time − Request Time

Example:

Request = 1.230 sec
Reply = 1.234 sec

RTT = 4 ms

Result

ICMP packets were captured successfully.

Explanation

ICMP helps verify network connectivity, detect failures, and measure delay using tools such as **ping** and **traceroute**.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

sol:

Aim

To configure a client-server network, access a web page, and analyze TCP and HTTP communication.

Topology

Client PC -------- Switch -------- Web Server

IP Configuration

Device	IP Address
Client PC	192.168.1.2

Web Server	192.168.1.10

Procedure

1. Place a PC and Server.
2. Connect through a switch.
3. Configure IP addresses.
4. Enable HTTP service on the server.
5. Open the client's web browser.
6. Enter:

   http://192.168.1.10

7. Start Wireshark capture.

TCP Three-Way Handshake

1. **SYN** – Client requests connection.
2. **SYN-ACK** – Server acknowledges and responds.
3. **ACK** – Client confirms the connection.

HTTP Communication

- Client sends **HTTP GET** request.
- Server responds with **HTTP Response (200 OK)** containing the webpage.

Result

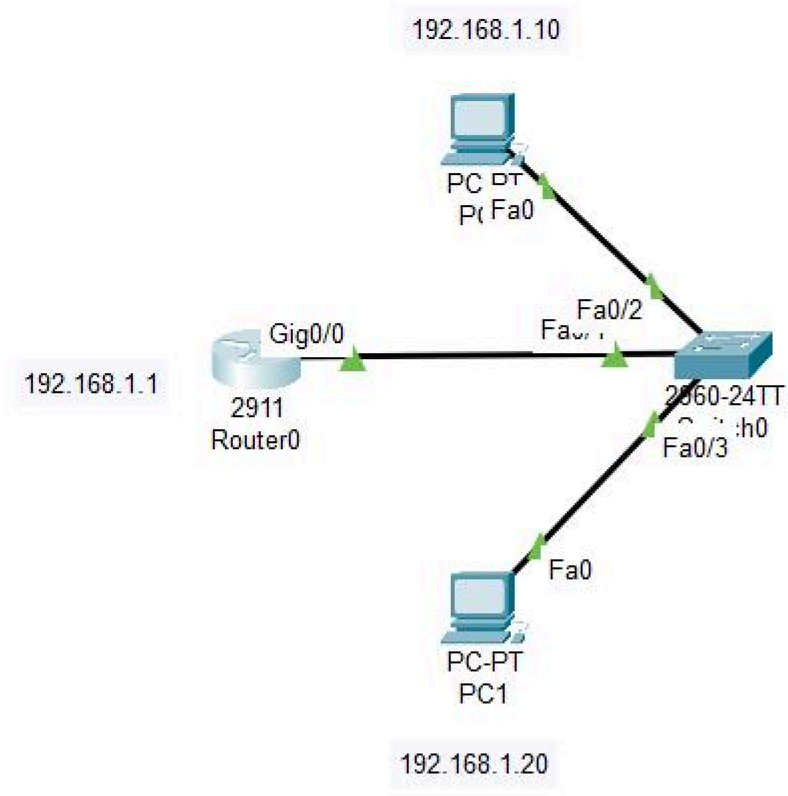
The webpage loaded successfully, and TCP/HTTP packets were captured.

Explanation

TCP provides reliable communication by:

- Establishing a connection with the three-way handshake.
- Using acknowledgments (ACKs).
- Retransmitting lost packets.
- Delivering data in the correct order.

HTTP uses TCP to ensure reliable transfer of web pages between the client and server.



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation